

Calculus I Lesson 19

Quotient Rule for Derivative: $D_x \left(\frac{f(x)}{g(x)} \right) = \frac{g(x) \cdot f'(x) - f(x) \cdot g'(x)}{[g(x)]^2}$

1) $h(x) = \frac{x^3 + 4}{x - 7}$

a) $D_x(x^3 + 4) = \underline{\hspace{2cm}} ?$

b) $D_x(x - 7) = \underline{\hspace{2cm}} ?$

c) $h'(x) = \underline{\hspace{2cm}} ?$

$$2) \ h(x) = \frac{x^2}{\sin x}$$

$$\text{a) } D_x(x^2) = \underline{\hspace{2cm}}?$$

$$\text{b) } D_x(\sin x) = \underline{\hspace{2cm}}?$$

$$\text{c) } h'(x) = \underline{\hspace{2cm}}?$$

$$3) \ h(x) = \frac{x^2 + 5x}{2x^2 + 5x - 5}$$

$$a) \ D_x(x^2 + 5x) = \underline{\hspace{1cm}}?$$

$$b) \ D_x(2x^2 + 5x - 5) = \underline{\hspace{1cm}}?$$

$$c) \ h'(x) = \underline{\hspace{1cm}}?$$

$$4) \ h(x) = \frac{\cos x}{5 + \sin x}$$

$$a) \ D_x(\cos x) = \underline{\hspace{2cm}}?$$

$$b) \ D_x(5 + \sin x) = \underline{\hspace{2cm}}?$$

$$c) \ h'(x) = \underline{\hspace{2cm}}?$$

$$5) \ h(x) = \frac{x^2}{\sin x}$$

$$\text{a) } D_x(x^2) = \underline{\hspace{1cm}}?$$

$$\text{b) } D_x(\sin x) = \underline{\hspace{1cm}}?$$

$$\text{c) } h'(x) = \underline{\hspace{1cm}}?$$

$$6) \ h(x) = \frac{x^{-2}}{2x^2 + \sin x}$$

$$a) \ D_x(x^{-2}) = \underline{\hspace{2cm}}?$$

$$b) \ D_x(2x^2 + \sin x) = \underline{\hspace{2cm}}?$$

$$c) \ h'(x) = \underline{\hspace{2cm}}?$$

$$7) \ h(x) = \frac{x^{2/3} + 1}{5x^2 + \sin x}$$

$$a) \ D_x(x^{2/3} + 1) = \underline{\hspace{2cm}}?$$

$$b) \ D_x(5x^2 + \sin x) = \underline{\hspace{2cm}}?$$

$$c) \ h'(x) = \underline{\hspace{2cm}}?$$

$$8) \ h(x) = \frac{x^{-3/4} + 7x}{x^3 + \cos x}$$

$$a) \ D_x(x^{-3/4} + 7x) = \underline{\hspace{2cm}}?$$

$$b) \ D_x(x^3 + \cos x) = \underline{\hspace{2cm}}?$$

$$c) \ h'(x) = \underline{\hspace{2cm}}?$$

$$9) \ h(x) = \frac{x^{-5/4} + 2x^3}{x^{1/2} + \cos x}$$

$$a) \ D_x(x^{-5/4} + 2x^3) = \underline{\hspace{1cm}}?$$

$$b) \ D_x(x^{1/2} + \cos x) = \underline{\hspace{1cm}}?$$

$$c) \ h'(x) = \underline{\hspace{1cm}}?$$

